DP Biology – D2.3 (HL extension)

Water potential

- What affects whether water moves into or out of a cell?
- In what ways do plant cells and animal cells control water movement differently?

HL statements D2.3.8–11. Textbook pp. 690–692.

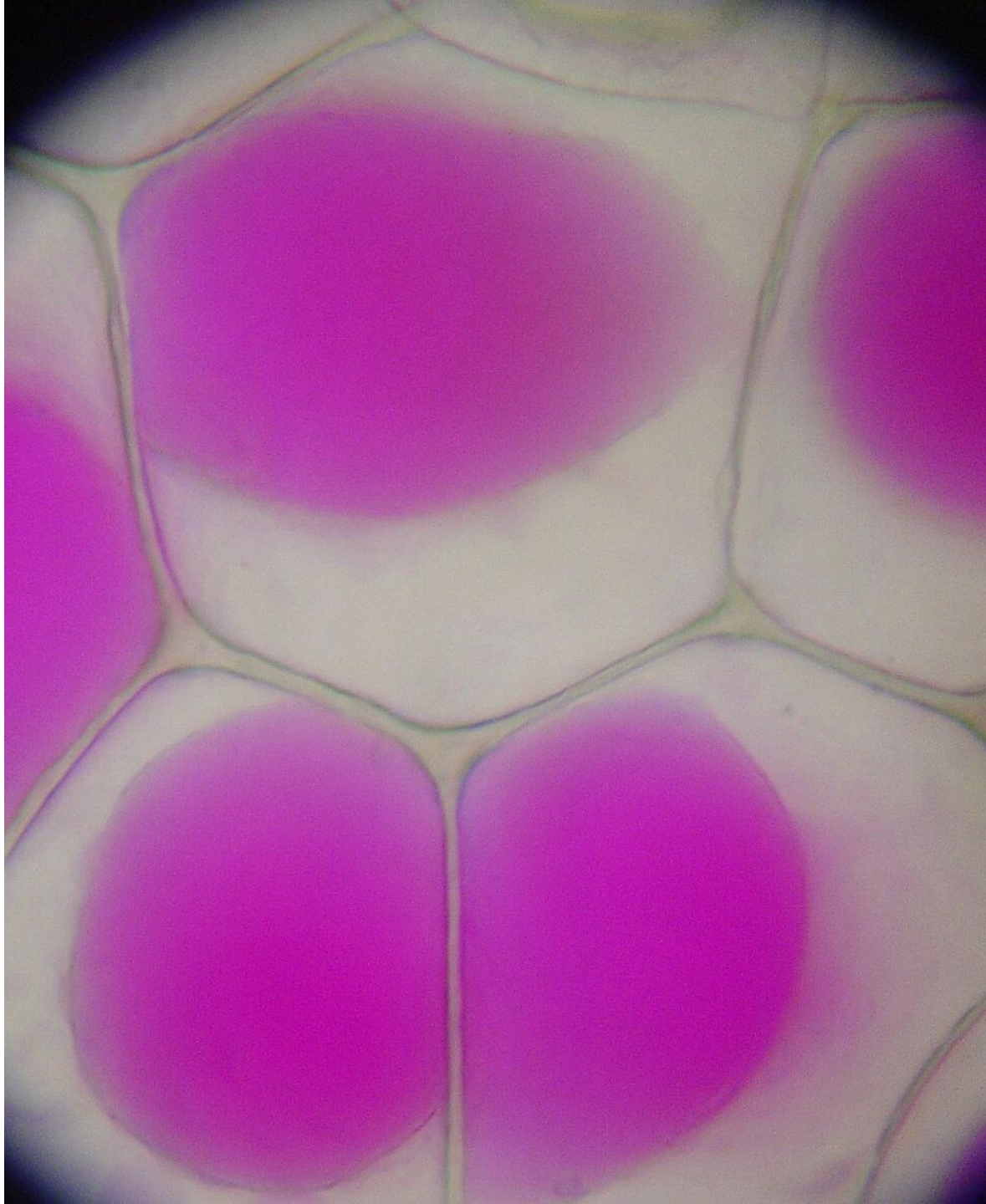


Image: plasmolysed cells of a leaf epidermis. Mnolf, CC BY-SA 3.0.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	D2.3.8–9	Define water potential and predict the direction of water movement
2	D2.3.10–11	Use solute and pressure potential to explain water movement in plant tissue

Why do plants wilt?



Roses wilted after a heat wave. CC0.

Water moves in and out of cells all the time. To predict **which way**, biologists use one number: **water potential**.

A wilting plant has lost so much water that its cells no longer press against their walls.

Question for today

How can one number tell us whether a cell will gain or lose water?

What is water potential?

Water potential (psi) tells us how much stored energy a given volume of water has, measured in kPa.

Pure water at standard pressure is set at **0 kPa**. Solutes lower it, so solutions are **negative**; the stronger the solution, the more negative.

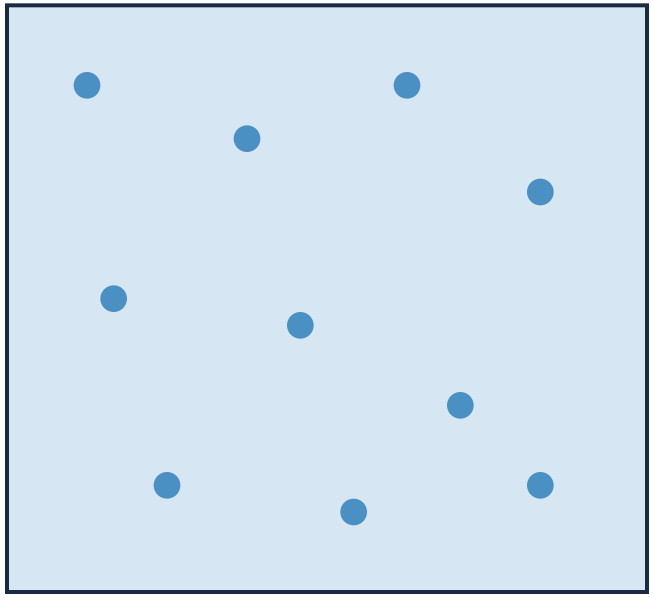
Liquid	psi / kPa
Pure water	0
Dilute solution	about -100
Leaf cell sap	about -1000
Sea water	about -2500

Units

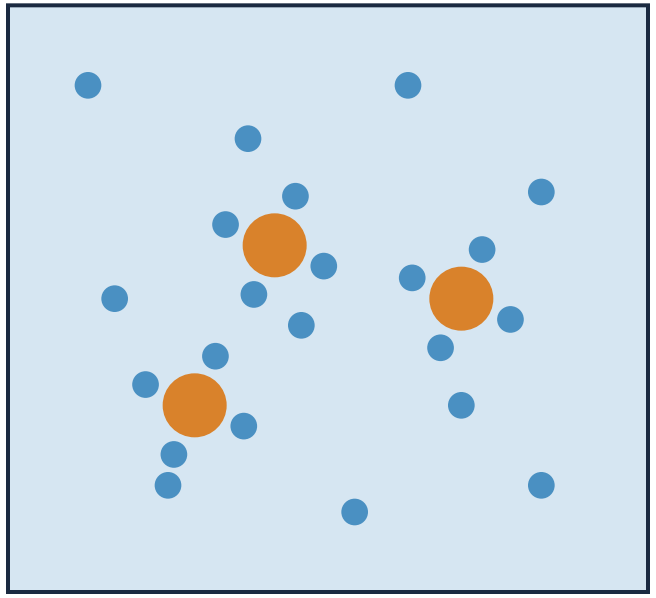
1 MPa = 1000 kPa. So -1000 kPa = -1.0 MPa.

Why solutes lower water potential

D2.3.8



pure water: 0 kPa



solution: negative

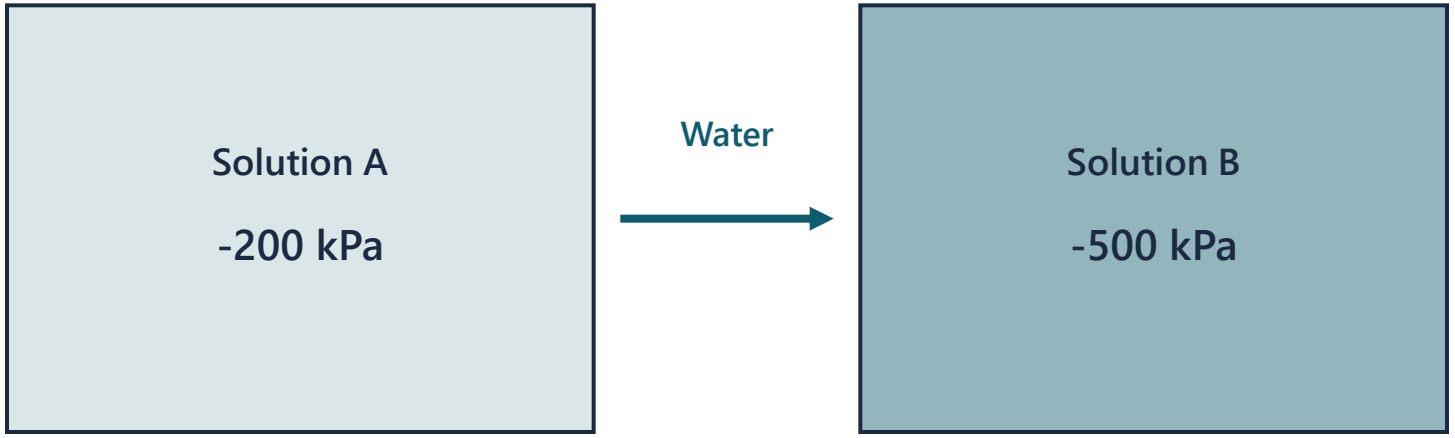
Drawn model. Blue: water molecules. Orange: solute particles.

Water molecules cluster around **solute** particles (orange), held by hydrogen bonds.

Fewer are free to move, so their potential energy is **lower**.

Water moves down a water potential gradient

D2.3.9



Separated by a partially permeable membrane. Darker: more solute.

Water moves from **higher** (less negative) to **lower** (more negative) water potential: here A to B.

It stops when the two are equal.

Key idea

This works for any pair: two solutions, a cell and a solution, or two cells side by side.

Practice: which way does water move?

D2.3.9

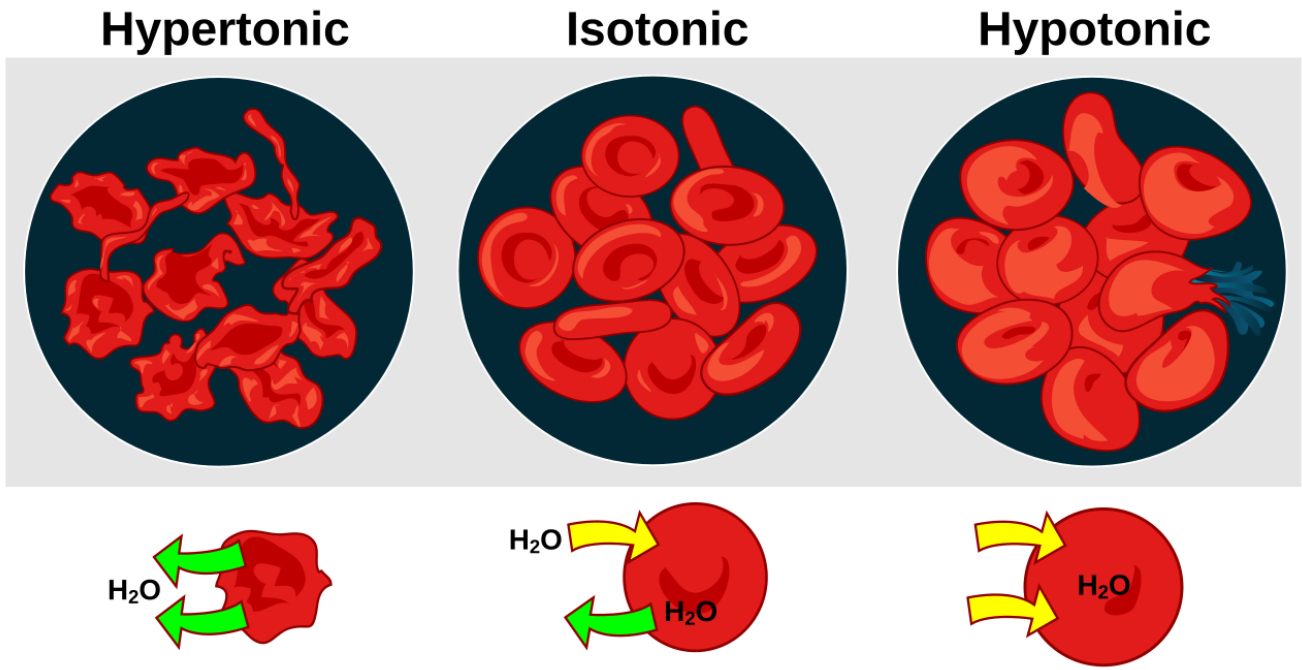
Cell X	Cell Y	Cell Z
-300 kPa	-500 kPa	-400 kPa

1. Draw arrows for net water movement between each pair.
2. Which cell gains most water?
3. Predict the final water potential of all three.

Hint

Compare each pair separately. Water always moves towards the more negative value.

Animal cells have no wall



Red blood cells in different solutions. LadyofHats, public domain.

Without a cell wall, no pressure can build up. Water potential depends only on **solutes**.

In a solution with a higher water potential, animal cells swell and can **burst**; in a lower one they **shrink**.

Link

This is why blood plasma and IV drips are kept at the same water potential as body cells.

Practice: units and direction

D2.3.8–9

Pair	Region 1	Region 2
A	-0.4 MPa	-650 kPa
B	0 kPa	-0.2 MPa
C	-1200 kPa	-1.2 MPa
D	-2.5 MPa	-900 kPa

1. Convert all values to kPa.
2. For each pair, state which way water moves.
3. In which pair is there no net movement?
4. Which pair has the steepest gradient?

Check-in: class 1

D2.3.8–9

1. Define water potential and state its unit.
2. State the water potential of pure water.
3. Explain why adding a solute lowers water potential.
4. State the direction of water movement between two regions.
5. Explain why animal cells can burst but plant cells do not.

How to use this

Answer without your notes, then check against slides 3–9.

Solute potential and pressure potential

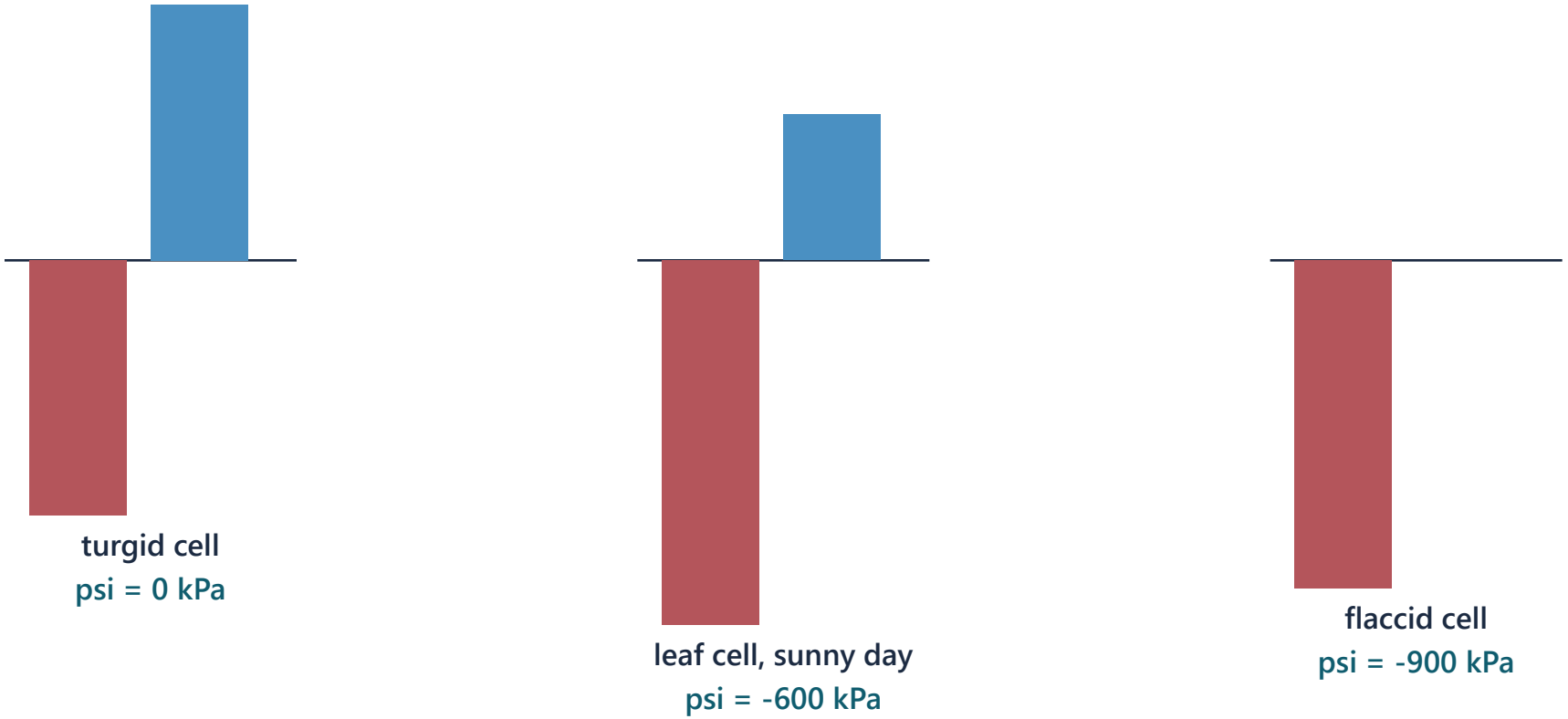
water potential = solute potential + pressure potential

	Solute potential	Pressure potential
Caused by	Dissolved solutes	Cell contents pressing on the wall
Usual sign	Zero or negative	Positive in turgid cells
Falls when	More solute dissolves	The cell loses water

Note
In xylem, sap is under tension, so pressure potential there is negative.

Seeing the equation

D2.3.10



Read the bars

Water potential = red bar (negative) + blue bar (positive). Taller blue bar = closer to zero.

Using the equation

D2.3.10

Cell	Solute potential / kPa	Pressure potential / kPa	Water potential / kPa
Turgid cell in pure water	−700	+700	0
Leaf cell, sunny day	-1000	+400	-600
Flaccid cell	-900	0	-900
Plasmolysed cell	-1200	0	-1200

Try it

Solute potential -800 kPa, pressure potential +300 kPa. Calculate the water potential.

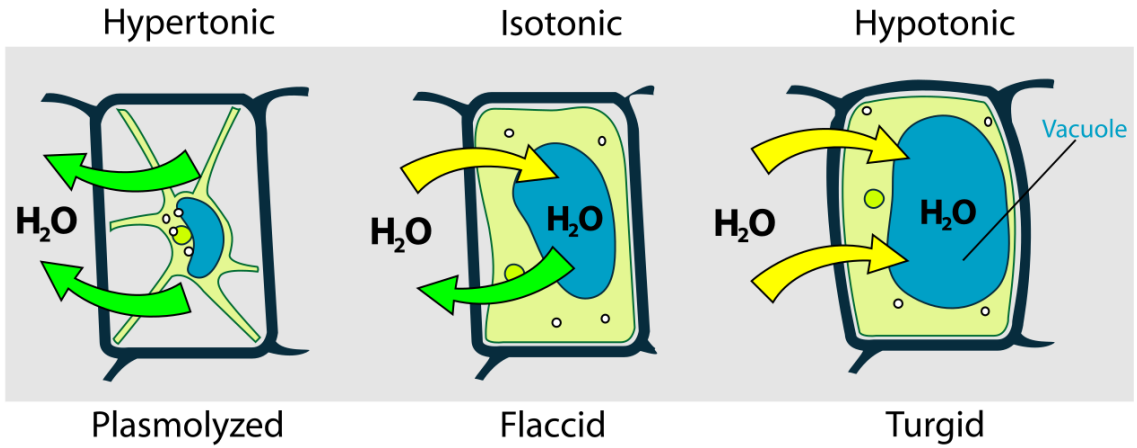
Then

It is put in a solution at -400 kPa. Does it gain or lose water?

Plant tissue in a hypotonic solution

D2.3.11

Solution psi higher than the cells → Water enters by osmosis → Solute potential less negative → **Pressure potential rises** → psi rises until equal



Plant cells in different solutions. LadyofHats, public domain.

Result

The cell becomes **turgid**. The wall stops it bursting. Turgid cells hold up leaves and young stems.

Plant tissue in a hypertonic solution

D2.3.11

Solution psi lower than the cells → Water leaves by osmosis → Solute potential more negative → Pressure potential falls to zero: flaccid → Protoplast pulls away: plasmolysis

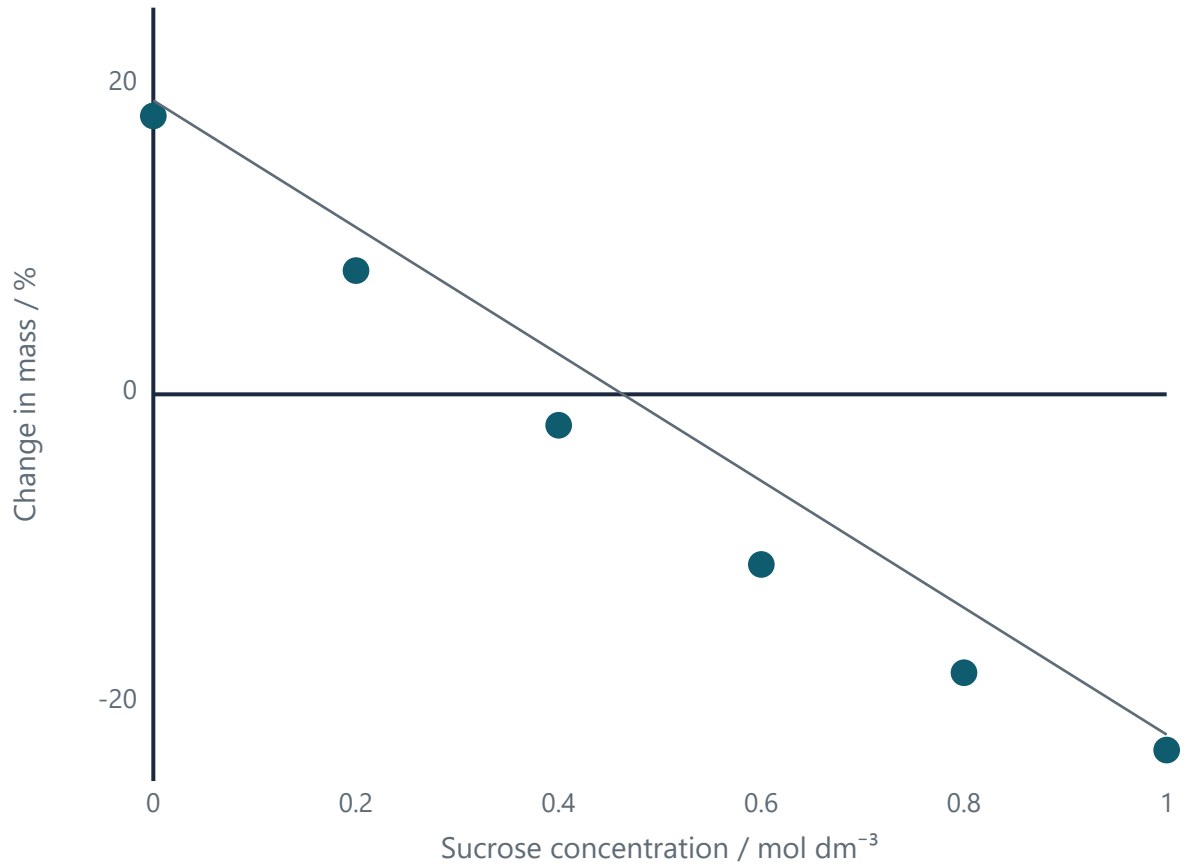


Plasmolysed epidermis cells. Mnolf, CC BY-SA 3.0.

Why plants wilt

In drought or salty soil, cells lose water and pressure potential falls, so leaves droop. Few plants survive in sea water, whose psi is far below that of most plant cells.

Estimating the water potential of a tissue



Invented results for potato cylinders after 24 h.

Weigh tissue samples, leave them in a range of sucrose solutions, then reweigh. Plot % change in mass.

Where the line crosses **zero**, there is no net water movement: the tissue has the **same** water potential as that solution.

Try it

Read the concentration where the line crosses zero.

Working with data: beetroot discs

D2.3.11

Percentage change in mass of beetroot discs in sucrose solutions. These data are invented for practice.

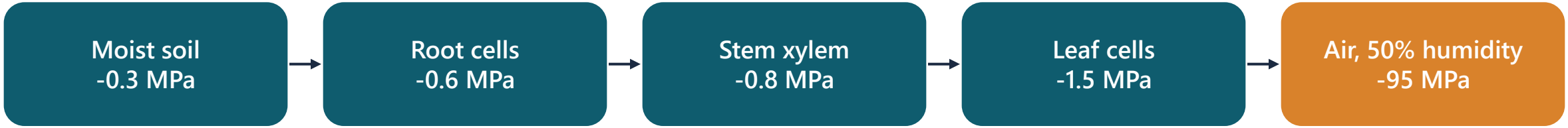
Sucrose / mol dm^{-3}	Solute potential / MPa	Mass change / %
0.1	-0.26	+8
0.2	-0.54	+3
0.3	-0.82	-2
0.4	-1.12	-7

1. Estimate the concentration with no mass change.
2. Estimate the water potential of the tissue.
3. Explain why repeats improve the estimate.

From soil to air

D2.3.8–10

Water potential falls at every step from soil to air. Typical values on a sunny day.



Think

Why is the biggest drop between the leaf and the air? What would happen if the soil dried to -2.0 MPa?

Practice: calculate and predict

D2.3.10–11

Cell	Solute / kPa	Pressure / kPa	psi / kPa
P	-1000	+600	?
Q	-800	?	-500
R	?	0	-1100

1. Fill in the gaps.
2. P and Q are side by side. Which way does water move?
3. Which cell is flaccid?

Check-in: class 2

D2.3.10–11

1. Write the equation linking the three potentials.
2. Explain how pressure potential changes as a plant cell takes in water.
3. Solute potential -0.8 MPa, pressure potential 0.3 MPa: calculate psi.
4. Describe what happens to a plant cell in a hypertonic solution.
5. Explain how to find the water potential of a tissue from sucrose solutions.

How to use this

Answer without your notes, then check against slides 11–19.

Common mistakes to avoid

Not this	But this
Water moves to a higher water potential.	It moves to a lower (more negative) one.
Solute potential can be positive.	Solute potential is zero or negative.
Pressure potential is negative in turgid cells.	It is positive in turgid cells.
-0.5 MPa is lower than -800 kPa.	-0.5 MPa = -500 kPa, which is higher.

Summary: water potential

D2.3.8–11

	Key idea
D2.3.8	Pure water 0 kPa; solutes make psi negative
D2.3.9	Water moves from higher to lower psi
D2.3.10	psi = solute potential + pressure potential
D2.3.11	Hypotonic: turgid. Hypertonic: flaccid, then plasmolysed

Check that you can...

1. **Define** water potential and **state** its units
2. **Explain** water movement using water potential
3. **Calculate** water potential from its components
4. **Explain** changes in plant tissue in different solutions
5. **Estimate** the water potential of a tissue from data

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. State the water potential of pure water.

2. A cell has a solute potential of -900 kPa and a pressure potential of +500 kPa. Calculate its water potential.

3. Explain what happens to the pressure potential of a cell placed in a hypertonic solution.

Image credits

Rhoeo Discolor – Plasmolysis.jpg – Mnolf – CC BY-SA 3.0

Roses wilted after heat wave at Gamla Strandgatan 11, Gamlestan, Lysekil.jpg – W.carter – CC0

Osmotic pressure on blood cells diagram.svg – LadyofHats – Public domain

Turgor pressure on plant cells diagram.svg – LadyofHats – Public domain

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